Energy & Buildings 175 (2018) 78–86

Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enbuild

Life cycle assessment of magnesium oxide structural insulated panels for a smart home in Vancouver

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ARTICLE INFO

Article history:
Received 11 January 2018
Revised 4 July 2018
Accepted 5 July 2018
Available online 18 July 2018

Keywords:
Life cycle assessment
LCA
Low energy house
Smart home
Structural insulated panel
SIP
Magnesium oxide
MgO
Stick-frame

ABSTRACT

Two major trends are changing the face of the construction industry. Sustainability concerns are driving innovations to improve building's energy efficiency while advances in prefabrication are improving buildings' speed, cost, and quality. As new materials and techniques emerge, their overall environmental impact must be evaluated using techniques such as life cycle assessment (LCA). This paper reports a life cycle assessment (LCA) of an innovative magnesium oxide structural insulated panel (MgO SIP) used for a high-performance smart home in Vancouver. The LCA compares the environmental impacts across six indicators for the MgO SIPs, traditional SIPs, and traditional stick-frame construction across the life cycle phases of raw material extraction, manufacturing, transportation, construction, and operation. The results show that the MgO SIPs do not outperform conventional alternatives notably due to the long-distance transportation of materials. However, further LCA of hypothetical scenarios demonstrates that MgO SIPs have a great potential to become more environmentally friendly than the conventional alternatives by sourcing MgO domestically, implementing local or onsite manufacturing, and designing the MgO SIPs without oriented strand board. Although the results are context-dependent, the investigation of hypothetical scenarios offers insight to improve the environmental performance of this innovative product.
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1. Introduction

Society's increasing concern for sustainability creates a demand for improved energy performance of homes, resulting in concepts such as low energy houses, zero energy houses and passive houses. Meanwhile, the building sector is realized its environmental impacts. One of the prevailing environmental analysis techniques is Life cycle assessment (LCA), which is an attempt to measure potential impacts of building products and components [1]. The ISO 14,040 standard states that LCA addresses the potential environmental impacts of products and services, both embodied and consumed, from extraction to final disposal [2] and it can help decision-making on product selection [3,4]. The European standard CEN/TC350 prescribes a life cycle approach for sustainable construction, including buildings [5] and products [6].

In an effort to understand the environmental impacts of high-performance houses, numerous studies have focused on life cycle energy analysis and life cycle CO_2 emissions, e.g. a life cycle energy analysis of a standard U.S. single-family house and a functionally equivalent energy-efficient house model [7], an analysis of primary energy use and CO_2 emission for the production and operation of 11 case study residential buildings with different types of energy supply systems in Sweden [8], a life cycle energy analysis of a typical Belgian passive house [9], and literature reviews of life cycle embodied energy of residential buildings [10,11]. Some research has taken into account comprehensive environmental impact indicators, e.g. a LCA of a low energy house in Italy [12] and a LCA of a stick-frame passive house with four different heating systems in Norway [13].

However, the majority of the studied houses were made of conventional materials, i.e., stick-frame, concrete, steel, and masonry; little research provides insights into innovative products such as structural insulated panel (SIP). SIPs consist of an insulating foam core sandwiched between two structural facings, typically oriented strand board (OSB). They are prefabricated under controlled factory conditions and then assembled on site. Life cycle benefits of SIPs are often claimed by manufacturers in the SIP industry, which could be subject to conflict of interest. For example, a comparative LCA of a single family house with SIPs and stick-frame showed that SIPs consume more embodied energy but save energy in operation phase, resulting in an average energy payback period of 2.7 years for Canada and 5.1 years for the U.S. [14]. Du et al. showed the

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https://doi.org/10.1016/j.enbuild.2018.07.016
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Fig. 1. Prototype house at the University of British Columbia Vancouver campus.

benefits of life cycle annual cost and annual CO_2 emissions for a SIP building compared to a masonry-concrete house [15]. Similarly, a comparative analysis of life cycle cost, embodied energy, and greenhouse gas emissions of SIP and stick-frame housing for U.S. military personnel proved lower operating cost and lower greenhouse gas emissions for SIP housing [16]. While few comprehensive LCAs have been reported in the literature for the common type of SIPs that are emerging within the housing industry, the focus of this study is on an even more innovative and uncommon product: a SIP clad in magnesium oxide (MgO) sheathing [17], which has the potential to outperform the alternatives of gypsum, OSB, plywood, and plastics in terms of its resistance to flame, water, mold, and insects. These MgO SIPs were used in a full-scale prototype house (Fig. 1) that was built on the University of British Columbia (UBC) Vancouver campus in late 2015.

2. Description of the prototype house

The prototype house was built by a start-up company based in Vancouver as a trial for a package of high-performance, low cost, "smart home" solutions. The prototype house is a two-storey single family house, with 104 m^2 ground floor area, 157.5 m^2 roof area (including the overhang), and 172.3 m^2 net exterior wall area (excluding windows and doors), adding up to 434 m^2 envelope. MgO SIPs were used for the majority of the envelope and structural systems (foundation walls, ground floor, exterior walls, and roof). Triple glazed windows and a high R-value building envelope were adopted as the passive design elements. The prototype house was also designed to reduce solar radiation in summer by the overhang (Fig. 2). The heating system is an air-to-water heat pump and the cooling is through natural ventilation. High-performance light emitting diode (LED) lighting elements and occupancy sensors to control the lighting are used where necessary. Controls throughout the house are managed by a building monitoring system (BMS), which also enables the data collection from temperature, relative humidity, carbon dioxide (CO_2), and volatile organic compounds (TVOC) sensors within the house in minutely intervals. Other advanced features include a grey water heat recovery system and the photovoltaic (PV) panels (5 kW) on the ground in front of the house.

3. Methodology

This LCA study follows the standard process provided by ISO 14,044 [2].

3.1. Goal

The goal of this LCA is to investigate the environmental impacts of a new product–MgO SIPs–used in a real case, in order to (1) inform the developer of the environmental impact of the chosen product for future decision-making, (2) provide insights to the industry by comparing the new product with traditional products, i.e., the traditional SIPs and the stick-frame system, and (3) suggest how to reduce the MgO SIP's environmental impact on future projects. This research focuses on the environmental impacts only, while other work has evaluated the economic, social, and technical performance of the prototype house as a research collaboration between the developer and UBC.

3.2. Functional Unit

According to ISO 14,040, "the functional unit defines the quantification of the identified functions (performance characteristics) of the product". In this LCA, all the alternatives were modelled to provide 434 m^2 of envelope for 50 years and the results are reported for 434 m^2·50 years.

The functions of a building envelope are water, air, vapour and heat control. Water control is typically accomplished by cladding (for example, siding, masonry or stucco) and a sheathing membrane or insulating sheathing, flashing, sealants and other materials. In Canadian wood-frame house construction, a combined air barrier and vapour barrier is often provided by 0.15 mm sheet polyethylene under a gypsum wallboard, sealed at joists, penetrations and interruptions, such as windows and floor junctions [18]. Rain screen systems are applied in some areas, and thermal insulation is always provided for heat flow control. In this study, we assumed all the alternatives had the same cladding, water proof sheathing membrane, and polyethylene sheet in the same locations, the same concrete foundation and timber structure elements outside of the studied panels, and the same doors and windows. Therefore, all the alternatives had the same level of controls over water, air, and vapour control. We only considered the thermal control function of the products in the LCA.

3.3. System boundaries

The phases considered in the LCA include the raw material extraction, manufacturing, transportation, onsite construction, and operation phase (Fig. 3). As a fairly new and uncommon product, the end-of-life phase recycle rate of the MgO SIPs is not known and thus the end-of-life phase was not modelled.

3.4. Life cycle inventory

A Life cycle inventory (LCI) is a collection of input and output data relating to the processes in the system studied. It often relies on an LCA database embedded in LCA tools. There are multiple LCA tools available in the market. The Athena Institute classified the assessment tools into three levels [19]: product comparison tools or information sources, whole building design or decision support tools, and whole building assessment frameworks or systems. Athena impact estimator for buildings (IE4B) is the only free LCA tool in North America for whole buildings and assemblies. Despite some arguments that an LCA of envelope system should be conducted within the context of a whole building instead of as a stand-alone product [20], GaBi 6 was selected to conduct a product LCA, because (1) the graph reports of IE4B do not meet the research needs; and (2) the MgO SIP panels, being an innovative product, are not available as a wall assembly option in IE4B.

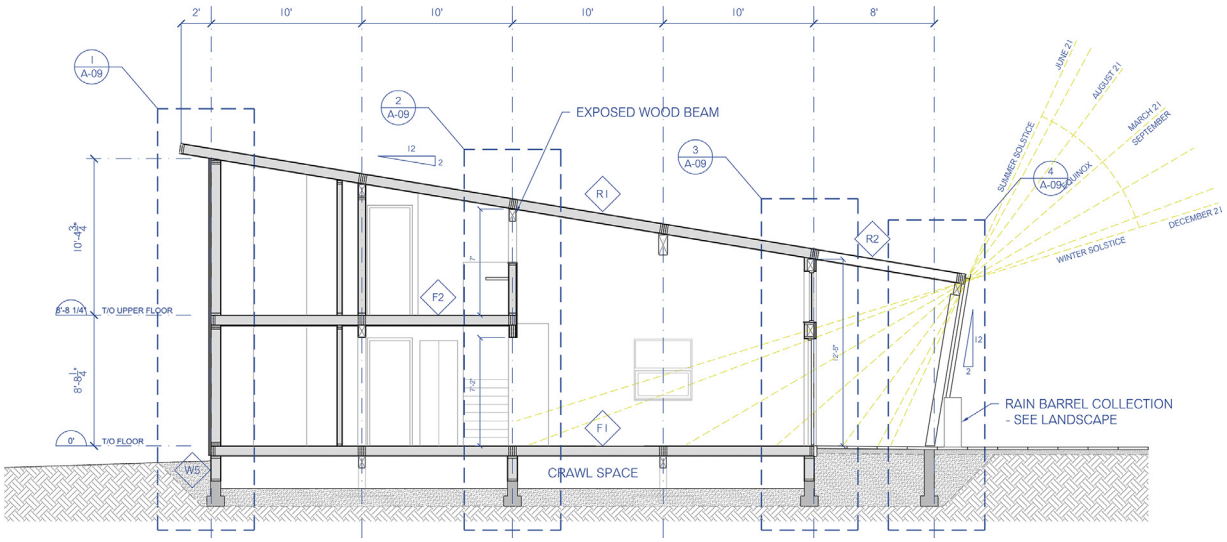


Fig. 2. North-south building section of the prototype house.

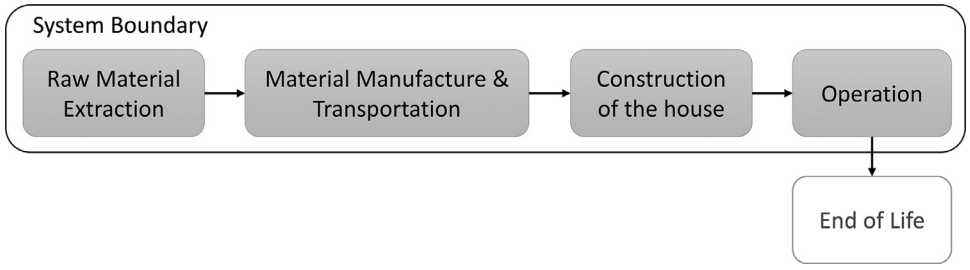


Fig. 3. System boundary of the LCA.

The amount of materials used for MgO SIPs was taken from the design drawings. The materials composition of the traditional SIP and stick-frame were determined based on assumptions, which are specified in Section 4. The material information of the MgO SIPs and traditional SIPs was used to define their manufacture process in GaBi. The material information of the stick-frame system was used to create its construction process in GaBi. The inventory datasets for other processes were mostly extracted from the GaBi database.

3.5. Life cycle impact assessment

Life cycle impact assessment (LCIA) translates the data in the LCI to potential environmental impact indicators through certain calculation rules. Table 1 reviews several widely used LCIA methodologies.

For this study, we are more interested in the chemical and physical changes in the environment (midpoints) rather than biological changes e.g. damage to human health (endpoints). Thus, we selected CML 2001, one of the earliest midpoint methods, as the LCIA methodology. Six environmental impact indicators were selected because they are common indicators and of the interest of the authors: Global Warming Potential (GWP), Acidification Potential (AP), Eutrophication Potential (EP), Ozone Layer Depletion Potential (ODP), Abiotic Depletion Fossil (ADF) and Human Toxicity Potential (HTP). The larger the indicator is, the more harmful it is.

4. Description of the LCA model

4.1. Materials of the alternatives

This LCA considered three alternative envelope products: MgO SIPs, traditional SIPs, and stick-frame. MgO SIPs were used in the real case, while traditional SIPs and stick-frame were hypothetical models based on industry norms with the principle of being comparable to MgO SIPs. Table 2 summarizes the materials that make up the assemblies for the three alternatives in LCA.

According to the structural drawings of the prototype house, the MgO SIPs consist of MgO sheathing, OSB, R-30.5 Expanded polystyrene (EPS), and softwood dimensional lumber (SDL) (2–2 × 6 studs at 24″ spacing with 2–2 × 8 top and bottom plates). The MgO SIP design is different for the exterior walls, floor, and roof. For traditional SIPs, OSB is used for sheathing instead of MgO, and gypsum wall board (GWB) is used for fire resistance. For the stick-frame case, we assumed 2–2 × 6 studs at 16″ spacing (giving 1.5 times the amount of dimensional lumber as with the SIPs) and fiberglass as the insulation, with one layer of 5/8″ plywood outside and one layer of 5/8″ GWB inside. Kellenberger and Althaus argue that the influence of the ancillary materials is high for wooden construction as a high quantity of screws, nails and other connectors are essential [28]. Therefore, screws and glue were also included in the LCA model. Screws were assumed to be 40 kg for MgO SIP and traditional SIP manufacturing and 80 kg for stick-frame construction.

For each alternative, we calculated the volume of each material by multiplying their thickness by 434 m^2 (the total area of envelope), and then calculated the mass of each material by multiplying their volume by their density. The mass of materials (Table 3) was

Table 1
LCIA methodologies.

LCIA methodologies	Created by	Description
CML 2001 [21]	Leiden University, the Netherlands	Provides characterization factors for more than 1700 different flows. The results are grouped in midpoint categories according to common mechanisms (e.g., climate change) or commonly accepted groupings (e.g., ecotoxicity).
Eco-indicator 99 [22]	PRé	Provides a single score for human health, ecosystem quality, and resources. The standard unit is point (Pt) or millipoint (mPt). The aim of this method is to compare products or components. The characterization factors for Eco-indicator 99 are also included in the CML 2001 documentation.
Ecological scarcity method 2013 [23]	ESU-services GmbH	The more the level of pollutant emissions or consumption of resources exceeds the environmental protection target set, the greater the eco-factor becomes, expressed in eco-points (EP). Just like Eco-indicator 99, the main aim of this method is to compare products and improve processes and/or products.
ILCD 2011 [24]	Joint Research Centre (JRC) of the European Commission	International Reference Life Cycle Data System recommends a method for each environmental theme, at both midpoint and endpoint.
ReCiPe 8 [25]	PRé, CML, RUN, and RIVM	The main objective of the ReCiPe method is to provide a method that combines Eco-Indicator 99 and CML, in an updated version. ReCiPe distinguishes midpoint indicators and endpoint indicators (damage to human health, damage to ecosystems and damage to resource availability).
TRACI 2.1 [26]	Environmental Protection Agency (EPA) of the U.S.	The tool for the reduction and assessment of chemical and other environmental impacts (TRACI) is to assist in LCIA, industrial ecology, and sustainability metrics. Methodologies were developed specifically for the U.S. using input parameters consistent with U.S. locations.
USEtox 2.0 [27]	UNEP/SETAC life cycle initiative	USEtox is a scientific consensus model endorsed by the UNEP/SETAC life cycle initiative for characterizing human toxicological and ecotoxicological impacts of chemical emissions in LCA.

Table 2
Composition of the three building envelope alternatives, exclusive of exterior cladding, roofing, etc. MgO = Magnesium Oxide board, EPS = Extruded polystyrene foam; OSB = Oriented strand board; GWB = Gypsum wall board; SDL = Softwood dimensional lumber.

	Prototype MgO SIP	Traditional SIP	Stick-frame
Exterior Wall	0.5″ MgO; 7.25″ EPS; 0.5″ OSB; 0.5″ MgO; 0.16″ glue; SDL; screws	0.5″ OSB; 7.25″ EPS; 0.5″ OSB; 0.625″ GWB; 0.16″ glue; SDL; screws	0.625″ Plywood; 7.25″ fiberglass insulation; 0.625″ GWB; SDL; screws
Floor	0.5″ MgO; 7.25″ EPS; 0.5″ OSB; 0.09″ glue; SDL; screws	0.5″ OSB; 7.25″ EPS; 0.5″ OSB; 0.625″ GWB; 0.09″ glue; SDL; screws	0.625″ Plywood; 7.25″ fiberglass insulation; 0.625″ GWB; SDL; screws
Roof	0.5″ OSB; 0.5″ MgO; 7.25″ EPS; 0.5″ OSB; 0.16″ glue; SDL; screws	0.5″ OSB; 7.25″ EPS; 0.5″ OSB; 0.625″ GWB; 0.16″ glue; SDL; screws	0.625″ Plywood; 7.25″ fiberglass insulation; 0.625″ GWB; SDL; screws

Table 3
Mass (kg) of materials in the three alternatives, acronyms same as in Table 2.

	MgO	OSB	EPS	SDL	Glue	Screws	GWB	Plywood	Fiberglass
MgO SIP	27,561	4806	1280	7921	1878	40	–	–	–
Traditional SIP	–	7053	1280	7921	1878	40	4670	–	–
Stick-frame	–	–	–	11882	–	80	4670	3736	2560

then treated as the input flows in the created manufacturing processes (for MgO SIPs and traditional SIPs) and construction process (for stick-frame) in GaBi.

4.2. Raw material extraction, manufacturing and transportation phase

According to the manufacturer, the MgO sheathing boards were produced in China, shipped to Vancouver, and then transported by rail to the factory. However, because we had no information regarding the production process of MgO sheathing boards such as the amount of energy used for production, in the LCA model, the "MgO production" process was created using MgO in China as the only input flow (available in GaBi). That is, the "MgO production" process in the LCA model only represented the raw material extraction in China. More discussion about the MgO source follows in Section 5.2. The raw materials of OSB, EPS, SDL, glue and screws were assumed to be supplied locally (160 km from the original site to SIP manufacturing factory via truck-trailer). The MgO SIPs were manufactured in a factory in High River, a town within the Calgary region of Alberta, Canada. After manufacturing, the MgO SIPs were transported from the factory to the construction site in Vancouver via rail.

For traditional SIPs, we assumed they were manufactured in Calgary as well and all the materials are from local suppliers (160 km via truck-trailer). After manufacturing, the traditional SIPs are transported from the factory in Calgary to the construction site in Vancouver via rail.

For the stick-frame, there is no manufacture process. All the materials were assumed to be locally sourced (160 km from Vancouver) and transported to the construction site by truck-trailer.

4.3. Construction phase

In the prototype house construction, when the MgO SIPs were delivered on site, the foundation walls were placed first on the



Fig. 4. MgO SIP panels being installed on the prototype house.

concrete strip footings, followed by the main floor panels. Then the wall panels were erected and connected to the floor. After columns and beams were installed, the roof panels were placed and connected to the walls. Fig. 4 shows the MgO SIPs being installed on site. Only one fork lift (with arm) was used to lift the panels during construction. The SIP installer said they used about 150 litres of diesel in total. In the LCA model, the construction process for MgO SIP case was modelled using 125 kg diesel input. The construction process of traditional SIP is the same as that of MgO SIPs.

To build a stick-frame house, SDL pieces firstly constitute the "frame", and then the plywood boards are connected to the outside of the frame, and then fiberglass is placed to the cavities in the frame, and the GWB is the interior finish. The construction of a stick-frame house clearly requires more manpower and time than SIP installation. However, according to the SIP installer, they would need only one fork lift as well for a stick-frame house with the same scale, and the use time of the fork lift is similar to that for SIP installation. Therefore, the energy used for construction was taken to be the same for all the three alternatives, which is 125 kg diesel input.

4.4. Operation phase

One of the major benefits of SIPs is the better thermal insulation than stick-frame. Therefore, we took into account the energy used for heating in the operation phase. Cooling is out of consideration because there is very limited need for cooling in Vancouver and the prototype house utilizes natural cooling. This LCA used the simulated electricity consumption as the input flow of the operation process, not the actual electricity consumption, because it is a prototype house that nobody lives in. At the time of the study, there was no valid performance data. The energy simulations were conducted by the developer company using HOT2000, which is an energy simulation and design tool for low-rise residential buildings developed by Natural Resources Canada [29]. The energy simulations for the SIP house and the stick-frame house were completed using the same software with the same assumptions for all the parameters (such as identical heat pump system) except the envelope type and R-value.

The average annual electricity usage for space heating in the prototype house was simulated to be 735 kWh. The electricity usage in the operation phase for a traditional SIP house would be the same because of the same envelope R-value. However, stick-frame assemblies can exhibit more than a 50% loss in R-value performance due to settling insulation, pinching, and degradation [30]. The 7.25″ fiberglass could provide a thermal insulation of approximately R-22. Considering the loss in R-value, for a conservative assumption, the R-value of the stick-frame house was taken to be R-20 in the energy simulation model. The simulation result for the stick-frame house was 1306 kWh for the average annual electricity usage for space heating.

The average annual electricity usage for space heating for each case was then multiplied by 50 years as the input flow of the operation process in this LCA. In Vancouver, more than 90% of the electricity is generated by hydro power, according to BC Hydro [31], the provider of electricity for British Columbia. Thus, we assumed the operational electricity all from hydro power in the LCA model. Discussion of this assumption follows in Section 6.1.

We did not include the solar energy in operation for the LCA, because the use of PV panels is independent from the adoption of MgO SIPs. The energy saving provided by PV panels should be consistent for all the alternatives, and thus this portion of saving has no need to be modelled as our purpose was product comparison.

5. Results

5.1. Comparison of the three alternatives

By comparing the LCA results of the three alternatives, it was found that the MgO SIPs envelope generally performed more poorly than traditional SIPs and stick-frame. An investigation of the main contributors to the environmental impacts (Table 4) revealed that the long-distance transportation and the processing of OSB were the main sources of the environmental impacts.

5.2. Scenario study for MgO SIPs

To further investigate the implications and opportunities of the MgO, we conducted LCAs on several MgO SIPs scenarios in order to identify strategies that could improve the environmental performance of the MgO SIPs. Table 5 summarizes the three new MgO SIP scenarios together with the original scenario.

Table 4
Main contributors to the environmental impact indicators.

	Prototype MgO SIP	Traditional SIP	Stick-frame
Global Warming Potential	Rail, ship, EPS	EPS, rail	fiberglass
Acidification Potential	Ship, OSB	OSB, rail	fiberglass
Eutrophication Potential	OSB, ship, rail	OSB, rail	fiberglass
Ozone Layer Depletion Potential	EPS	EPS	fiberglass
Abiotic Depletion Fossil	EPS, ship, transportation (diesel)	EPS, transportation (diesel)	fiberglass, diesel
Human Toxicity Potential	EPS, diesel, OSB	EPS, diesel, OSB	plywood processing

Table 5
Settings of the MgO SIP scenarios.

	Prototype MgO SIP	MgO SIP scenario A	MgO SIP scenario B	MgO SIP scenario C
MgO Source	China	Canada	Canada	Canada
Manufacturing site	Calgary	Calgary	Vancouver	Vancouver
MgO SIP design	With OSB	With OSB	With OSB	Without OSB

First, it was found that the transportation of MgO from China to Canada contributed significantly to the GWP, AP and EP. As deposits of MgO ore do exist in Canada, another scenario ("MgO SIP scenario A") was evaluated to consider the case where the MgO sheathing boards were supplied from Canadian source and transported to the factory in Calgary via rail (800 km). There are questions regarding this feasibility and impacts of sourcing MgO and producing the MgO board in Canada, but these are out-of-scope for this study. Since the environmental impact of the MgO production in China was not modelled in the initial LCA (explained in Section 4.2), this scenario becomes a fair comparison with the prototype MgO SIP and reflects the exclusive effect of long-distance ship transportation.

Second, the transportation between factory and construction site was also a significant contributor. Therefore, we considered the concept of micro-manufacturing and created a scenario named "MgO SIP scenario B" where all the materials including MgO were locally sourced and delivered directly to the construction site, then the MgO SIPs were manufactured in small onsite facilities, and then the MgO SIPs were installed to build the house as usual (similar results are assumed if the MgO SIPs were manufactured off-site but local to the construction site)

Finally, the OSB was initially used in the MgO SIP to meet seismic design requirements. Later, a new design of MgO SIP without OSB passed seismic testing, according to the research conducted by the developer company. Therefore, we analysed another scenario named "MgO SIP scenario C" where the MgO SIPs consist of ½" MgO board, 7.25″ EPS insulation, and ½" MgO board for walls, roof and ground floor, without OSB; the MgO are sourced in Canada; and the MgO SIPs are manufactured locally in Vancouver.

Fig. 5 shows the LCA results of the total six scenarios. The real case "Prototype MgO SIP" is marked as black.

Comparing the black bar to the bars on its left (the two traditional alternatives), it is found that the Prototype MgO SIP has the highest impact on GWP, AP and ADF. The stick-frame has the lowest impact in terms of GWP, AP, EP and ADF, but it has considerably higher impact on ODP and HTP because the fiberglass in the stick-frame has a greater impact than EPS in terms of ODP, and the processing of plywood causes a large impact on HTP.

However, looking from the black bar to its right, we found that the three strategies could make MgO SIPs more environmentally friendly than the conventional alternatives. By sourcing MgO from Canada ("MgO SIP scenario A") rather than China, MgO SIPs would have similar or lower impacts than traditional SIPs. If onsite manufacturing was also used ("MgO SIP scenario B"), the MgO SIPs would be significantly better than traditional SIPs because the elimination of transportation between the factory and site reduces the impact on most of the indicators. Moreover, if three strategies were applied together—that is, the "MgO SIP scenario C"—MgO SIPs would perform better than a stick-frame house for all indicators except for ADF.

The preliminary conclusion is that the MgO SIPs used in the prototype house do not environmentally outperform the conventional alternatives. However, they have the potential to beat the conventional alternatives if sourcing MgO locally, applying onsite or local manufacturing, and eliminating OSB from SIPs.

6. Discussion

6.1. The effect of electricity source

The results above show that applying the three strategies to MgO SIPs still cannot make it superior than stick-frame in terms of ADF. As mentioned earlier, more than 90% of the electricity in Vancouver comes from hydro power, which has relatively low environmental impact. If the house uses electricity generated from coal or natural gas, the results may be different. Therefore, we created two arbitrary scenarios where the electricity was entirely from natural gas or hard coal for both "MgO SIP scenario C" (three strategies applied together) and "stick-frame" . As Fig. 6 shows, if the electricity is from natural gas or hard coal, the MgO SIP scenario C will be superior to stick-frame envelope in terms of ADF. This is because SIPs require less energy in the operation phase. When the electricity in the 50-year operation becomes a major contributor to the environmental impact, this benefit of MgO SIPs stands out. A previous study proved the effect of electricity source by studying the life cycle impact of the same residential heating and cooling systems in four regions in the US (considering the real electricity grid mix of the four regions) [32].

6.2. Limitations

This life cycle inventory analysis (GaBi database) relies on the conventional process analysis, which is known to suffer from "truncation error" which can greatly underestimate the impacts [33–36]. The truncation error is due to the setting of system boundaries and thus the omission of processes outside these boundaries. To solve this problem, researchers suggest complementing the process analysis with an input-output analysis covering all omissions, thus resulting in a hybrid LCA method. In their studies on whole buildings, Crawford [37], Crawford and Stephan [38], and Stephan and Stephan [39] have shown that input-output-based hybrid analysis can produce embodied energy figures about four times higher than a process analysis for the same building. Similarly, Wiedmann et al. [40] studied wind turbines in the UK using process and hybrid analysis and found that hybrid analysis

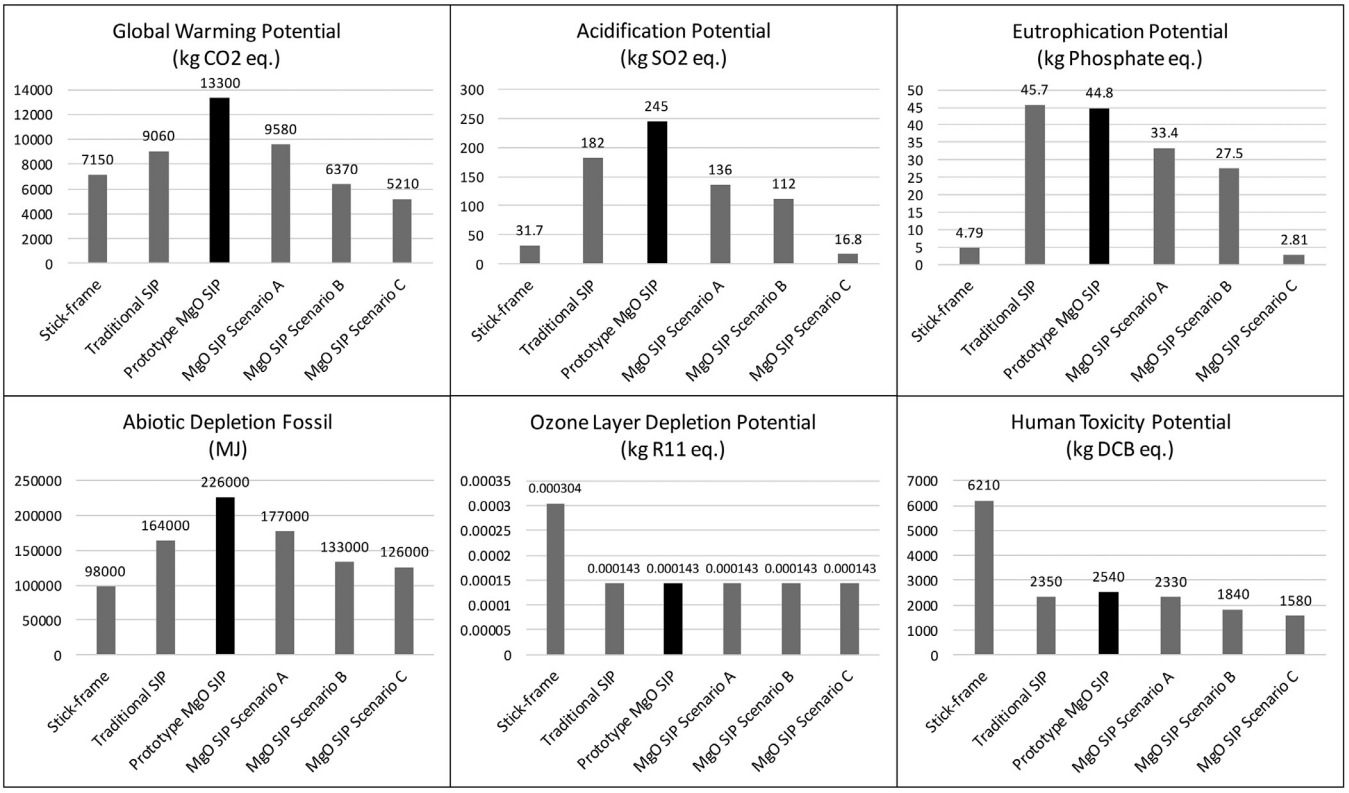


Fig. 5. LCA results of the six scenarios. Settings of MgO SIP scenarios A, B, and C are in Table 5.

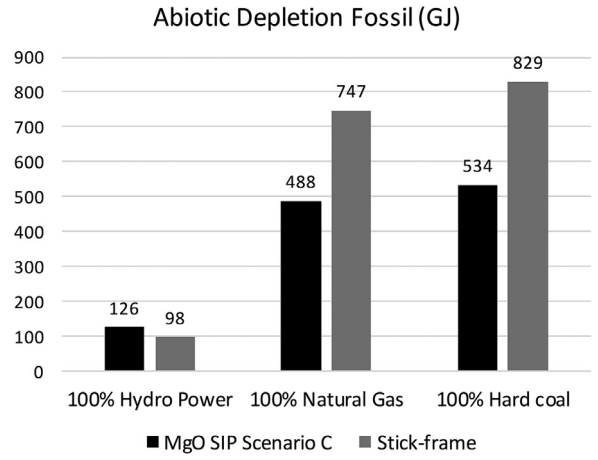


Fig. 6. Comparison of MgO SIP scenario C (no OSB, MgO from Canada, and manufactured onsite) and stick-frame with different electricity sources.

resulted in environmental impacts two times higher than when using process data. This means that the ratio of embodied to operational energy is probably higher than what we found in our study, along with the overall impact of each scenario. Nevertheless, it is important to note that the comparative outcomes of the alternatives and scenarios (i.e., the ranking of their environmental impacts) are very likely to remain valid.

Another limitation is the neglected SIPs benefit of structure support. Usually, the envelope is not responsible for structure support, but SIPs are load bearing, reducing or removing the need for posts and beams. However, in the prototype house, there still exist some posts and beams for structure support. At the time of the study, there was no structure analysis available to prove that the MgO SIPs alone could provide enough structure support, therefore, we decided to accept the actual design and did not include the structure support function in our LCA, meaning that the post and beam structure were excluded for all of the alternatives. If the SIP's benefit of structure support was proved and taken into consideration in the LCA model as a function, we would have to add posts and beams to the stick-frame case to generate a fair comparison, resulting in higher environmental impacts for the stick-frame.

6.3. Future research

Future LCA should incorporate the end-of-life phase. SIPs have a great potential for reuse, for instance, by disassembling the house and rebuilding in another place, while stick-frame house assemblies do not generally achieve a high degree of material reuse. A cradle-to-grave whole LCA study can be conducted once the end-of-life phase recycle rate of MgO is known.

In addition, this study analysed environmental impacts only, but a future economic analysis for MgO SIP, traditional SIP, and stick-frame, and for the new MgO scenarios, could provide valuable information for product selection. This potential to develop Canadian-sourced MgO panelling, in particular, has extensive economic and social implications. In addition, the fast construction enabled by SIPs can reduce the construction cost (labour cost, equipment and materials cost, etc.) to a large extent. Operation energy saving is also an economic factor favourable for SIPs. Moreover, among SIPs technologies, MgO has the potential to outperform the alternatives of gypsum, OSB, plywood, and plastics in terms of its resistance to flame, water, mold, and insects. These features may reduce the need of other cladding of the building envelope, translating to potential cost saving. On the other hand, the cost of MgO could be higher than conventional materials. Therefore, an economic analysis is necessary to validate the new product's economic benefit, if exits.

Another potential future direction is to analyse the effect of house size on the adoption of the MgO SIPs. Would reducing the house size result in more significant energy saving and thus magnify the benefits of SIPs? This is quite relevant for the Canadian context where houses are some of the largest in the world (in terms of floor area per capita). Stephan and Crawford [41] argue that as the gross floor area reduces, the life cycle energy demand per m^2 increases, while the life cycle energy demand per capita decreases.

7. Conclusions

The building industry is moving towards prefabrication, including the use of SIPs because of their benefits in terms of quality control, improved thermal insulation, operational energy saving, fast construction, and winter construction. Among SIP technologies, MgO has recently been introduced as a sheathing alternative to OSB because MgO is resistant to flame, water, mold, and insects. To inform the industry of the environmental impacts of the new product, we conducted a LCA of MgO SIPs, comparing a real house built with MgO SIPs with hypothetical similar houses built with traditional SIPs and stick-frame construction.

The LCA results show that the MgO SIPs do not perform better than the conventional alternatives in terms of environmental impacts, mainly because of the long-distance transportation of materials and the production of OSB. By conducting further LCA for hypothetical scenarios, we conclude that the MgO SIP could potentially show environmental advantages relative to the conventional alternatives through domestic sourcing of MgO, onsite manufacturing, and the elimination of OSB.

However, for many of the environmental impact indicators, the differences between MgO SIP, traditional SIP and stick-frame are not large and are quite dependent upon assumptions relating to the transportation of materials, etc. Moreover, if the SIP house is located in another place where electricity is generated from hard coal or natural gas, SIP will be significantly more sustainable than stick-frame because of the operational energy saving arising from the improved insulation. The potential end-of-life reuse of MgO SIPs could lead to an additional large benefit; however, this was not modelled in the LCA due to unknown recycle rate. The structural support benefit of SIPs was also intentionally omitted in this LCA because the prototype house utilizes posts and beams along with SIPs. The process analysis used could result in the estimation lower than actual environmental impacts. However, this should not affect the comparative results.

Since this LCA is based on a specific case and it is found that the LCA is so sensitive to the context and design parameters, the results should not be generalized, yet they may contribute along with many other qualitative and quantitative factors in informing design solutions. Future research could involve a cradle-to-grave LCA, life cycle costing of MgO SIPs, and evaluation of the factors that could influence the adoption of MgO SIP.

Declarations of interest

None.

Acknowledgement

We gratefully acknowledge the active support and collaboration of AYO Smart Homes as an industrial partner of this research, the financial support from the Natural Sciences and Engineering Research Council of Canada, the UBC Sustainable Building Science Program, and the UBC Centre for Interactive Research on Sustainability, and the software support from GaBi Education group.

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